overload occurs. This defect cannot be found even by multimeter testing. It can be rectified only by replacing the existing transformer with a new one. Partial short-circuit may result in minor fire accidents. Partial short- circuit may occur in Line Output Transformer (LOT), which is used in CRT model television circuit. Figure 1.17 shows an example for partial short-circuit.

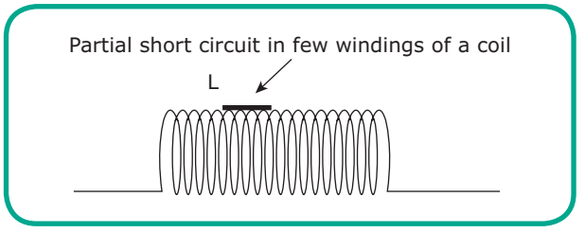


Figure 1.17 Coil showing partial short-circuit

1.4.13 Series Circuit

A circuit which has current flow in only one path through more than one load connected in serial manner (one after another load) with a power supply is said to be series circuit. Figure 1.18 shows a series circuit.

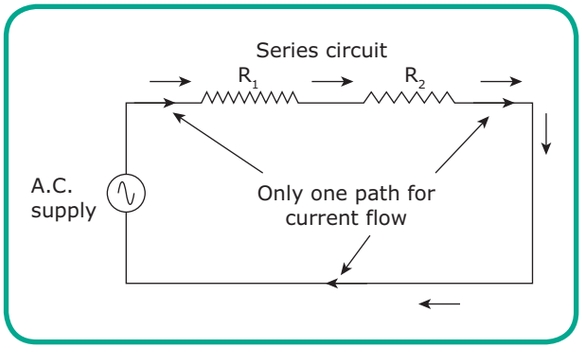


Figure 1.18 Series circuit

1.4.14 Parallel Circuit

A circuit, in which all the loads connected, have individual path for current flow, then it is said to be parallel circuit. Figure 1.19 shows the parallel circuit.

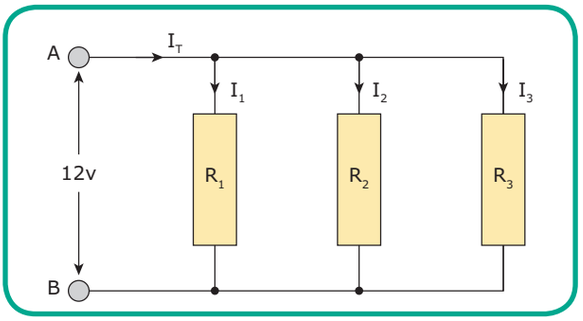


Figure 1.19 Parallel circuit

1.5 RESISTORS CONNECTED IN SERIES

Figure 1.20 shows three resistors R_1, R2 and R_3connected in series. When the voltage (V) is applied to the circuit, the current flow through the circuit and the following important points are to be noted.

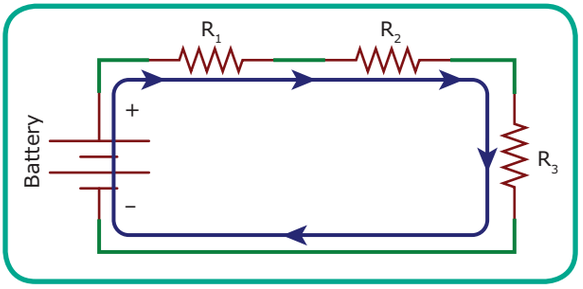


Figure 1.20 Resistors in series

1. The current I is equal in all the resistors.
2. The voltage drop across each resistor depends upon their resistance value i.e.,

$$V_1 = I.\ R$$
$$V_2 = I.\ R$$
$$V_3 = I.\ R$$

3. The EMF applied to the circuit is equal to the sum of the voltage drops across all resistors (Kirchhoff's voltage law). Hence $V = V_1 + V_2 + V_3$
We can derive formula for Total resistance of a series circuit from the above equation.

ELEC_EQUIP_Chapter01.indd 12 11/10/2020 12:22:40 PM